IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

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DECLARATION

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ABSTRACT

This research presents the design and development of an IoT-Enabled Smart Aquaculture Management System that integrates multiple intelligent modules to improve efficiency, sustainability, and decision-making in small and medium-scale fish farming environments. The system combines four key functional components: an Automated Feeding System, a Fish Mortality Detection Module, a Smart Water Quality Monitoring and Maintenance System, and a Budget Tracking and Cost Analytics Module.

Traditional aquaculture practices rely heavily on manual observation, fixed feeding schedules, and fragmented record-keeping methods. These approaches often lead to inefficient resource utilization, delayed detection of critical conditions, and increased operational costs. To address these limitations, the proposed system leverages Internet of Things technologies, computer vision techniques, and cloud-based analytics to provide real-time monitoring, automation, and intelligent decision support.

The Automated Feeding System ensures precise feed distribution using weight-based control, reducing waste and improving feeding consistency. The Fish Mortality Detection module utilizes image processing and machine learning techniques to identify abnormal fish behavior and detect mortality events in real time. The Water Quality Monitoring module continuously tracks key environmental parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and carbon dioxide, while incorporating condition-based maintenance strategies to generate actionable maintenance tasks. The Budget Analytics module integrates energy monitoring and expense tracking to provide cost analysis, forecasting, and financial insights.

The system is implemented using ESP32-based edge devices, MQTT communication via EMQX, real-time cloud services, and a web-based dashboard for visualization and control. Experimental results demonstrate that the system provides accurate monitoring, reliable data transmission, and meaningful insights that support efficient farm management.

Overall, the proposed solution offers a low-cost, scalable, and integrated platform that enhances productivity, reduces risks, and promotes data-driven aquaculture practices.

Keywords: Aquaculture, Automated Feeding System, Fish Mortality Detection, Internet of Things (IoT), Water Quality Monitoring.

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LIST OF ABBREVIATIONS

IoT	Internet of Things
MQTT	Message Queuing Telemetry Transport
CBM	Condition-Based Maintenance
DO	Dissolved Oxygen
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
VPS	Virtual Private Server
API	Application Programming Interface
RPA	Robotic Process Automation
MAE	Mean Absolute Error
ARIMA	Auto Regressive Integrated Moving Average

1 INTRODUCTION

1.1 Background of the Study

Aquaculture has become one of the fastest-growing food production sectors worldwide, playing a significant role in ensuring food security and supporting economic development. In countries such as Sri Lanka, small and medium-scale fish farming is widely practiced, contributing to both local consumption and income generation. However, managing aquaculture operations effectively remains a complex challenge due to the dynamic nature of aquatic environments and the reliance on manual processes.

Traditional aquaculture practices depend heavily on human observation and experience. Feeding is typically performed manually based on fixed schedules, which can result in overfeeding or underfeeding. Similarly, water quality monitoring is conducted periodically rather than continuously, making it difficult to detect sudden environmental changes. Fish health assessment, including mortality detection, is also largely dependent on visual inspection, which may delay the identification of critical issues.

In addition to operational challenges, financial management is often handled using basic record-keeping methods such as notebooks or spreadsheets. This fragmented approach limits the ability of farmers to analyze costs, track resource usage, and make informed decisions. As a result, inefficiencies in feeding, water management, and energy consumption can lead to increased operational costs and reduced productivity.

Recent advancements in Internet of Things technologies have introduced new opportunities to modernize aquaculture systems. IoT enables the integration of sensors, embedded devices, and cloud platforms to collect and analyze data in real time. These technologies allow continuous monitoring of environmental parameters, automation of key processes, and remote access to system information.

However, most existing solutions focus on individual aspects such as water quality monitoring or basic automation. There is a lack of integrated systems that combine multiple functionalities, including feeding automation, fish health monitoring, environmental analysis, and cost management, into a single platform.

To address these limitations, this research proposes an IoT-Enabled Smart Aquaculture Management System that integrates multiple modules into a unified solution. By combining automation, real-time monitoring, intelligent analysis, and financial tracking, the system aims to improve efficiency, reduce risks, and support data-driven decision-making in aquaculture operations.

1.1 Problem Statement

Small and medium-scale aquaculture farms face several challenges due to the absence of integrated and intelligent management systems. Feeding practices are often inconsistent, leading to feed wastage or insufficient nutrition for fish. Water quality is monitored manually and infrequently, making it difficult to detect sudden environmental changes that may affect fish health.

Fish mortality detection is another critical issue, as farmers rely on visual observation, which can delay response time and increase losses. Additionally, operational costs such as electricity consumption, feed expenses, and maintenance are not monitored in real time, resulting in poor financial planning and inefficient resource utilization.

Although IoT-based systems have been introduced in aquaculture, most existing solutions address only individual aspects and do not provide a comprehensive platform that integrates automation, monitoring, analytics, and decision support.

Therefore, the main problem addressed in this research is the lack of a low-cost, integrated, and intelligent system that can:

- Automate feeding operations accurately
- Detect fish mortality in real time
- Continuously monitor water quality parameters
- Provide intelligent maintenance recommendations
- Track and analyze operational costs
- Support decision-making through real-time insights

1.2 Research Objectives

1.2.1 Main Objective

To design and develop an IoT-based Smart Aquaculture Management System that integrates feeding automation, fish mortality detection, water quality monitoring, and cost analytics into a unified platform.

1.2.2 Specific Objectives

- To develop an automated feeding mechanism with precise feed control
- To implement a fish mortality detection system using computer vision techniques
- To monitor water quality parameters in real time using IoT sensors
- To implement condition-based maintenance for water management
- To design a cost tracking and analytics module for financial management
- To integrate all modules into a cloud-based system with real-time dashboard
- To provide alerts and decision support for efficient farm management

1.3 Scope of the Study

This research focuses on the design and implementation of an integrated IoT-based aquaculture system suitable for small and medium-scale fish farms. The system includes four main modules: automated feeding, fish mortality detection, water quality monitoring, and budget analytics.

The study covers real-time data collection using sensors and cameras, cloud-based data processing, and user interaction through a web dashboard. The implementation is limited to a prototype-level system and does not include large-scale industrial deployment.

1.4 Significance of the Study

This study provides a comprehensive solution for improving aquaculture management by integrating multiple technologies into a single system. The proposed system enhances operational efficiency by automating feeding, improving water quality monitoring, and enabling early detection of fish mortality.

The inclusion of financial analytics allows farmers to track expenses, analyze costs, and make better budgeting decisions. By providing real-time insights and intelligent recommendations, the system supports data-driven decision-making.

Furthermore, the low-cost design ensures accessibility for small-scale farmers, promoting the adoption of smart technologies in aquaculture and contributing to sustainable farming practices.

2 LITERATURE REVIEW

2.1 Introduction

The rapid growth of aquaculture has increased the demand for efficient, reliable, and intelligent management systems capable of supporting sustainable fish farming practices. As aquaculture environments are highly dynamic and sensitive to multiple factors such as feeding patterns, water quality, and fish health conditions, traditional manual methods are no longer sufficient to ensure optimal productivity. Recent advancements in digital technologies, particularly the Internet of Things, machine learning, and real-time analytics, have opened new possibilities for improving monitoring, automation, and decision-making in aquaculture systems.

This chapter reviews existing research and technological developments related to smart aquaculture systems. The discussion focuses on four key areas relevant to this study: automated feeding systems, fish mortality detection using computer vision, water quality monitoring with intelligent maintenance strategies, and cost analytics with predictive forecasting. In addition, the role of IoT communication, cloud integration, and real-time data processing is examined. The aim is to identify the strengths and limitations of current solutions and highlight the need for a fully integrated system that combines these functionalities into a single platform.

2.2 IoT-Based Aquaculture Monitoring Systems

The adoption of Internet of Things technologies has significantly transformed aquaculture by enabling continuous monitoring and remote management of fish farming environments. IoT-based systems typically consist of sensors, microcontrollers, communication protocols, and cloud-based platforms that work together to collect, transmit, and analyze data in real time. These systems allow farmers to monitor key environmental parameters without being physically present at the farm, thereby improving efficiency and reducing manual effort.

Several studies have demonstrated the effectiveness of IoT in monitoring water quality parameters such as temperature, pH, dissolved oxygen, and turbidity. These parameters play a critical role in maintaining a healthy aquatic environment, and continuous monitoring helps prevent sudden changes that could harm fish populations. IoT systems provide real-time alerts when parameter values exceed safe thresholds, enabling timely intervention.

Despite these advantages, most existing IoT-based aquaculture systems are designed as standalone solutions that focus on a single aspect of farm management. While they provide valuable environmental data, they often lack integration with other important components such as feeding automation, fish health monitoring, and financial analysis. As a result, farmers are required to use multiple systems or rely on manual processes to manage different aspects of their operations, which reduces overall efficiency.

2.3 Automated Feeding Systems in Aquaculture

Feeding is one of the most critical activities in aquaculture, directly influencing fish growth, health, and production efficiency. Traditional feeding methods rely on manual distribution based on fixed schedules or farmer experience. This approach often leads to inconsistencies, resulting in overfeeding or underfeeding. Overfeeding increases operational costs and contributes to water pollution, while underfeeding affects fish growth and productivity.

Automated feeding systems have been introduced to address these challenges by providing controlled and consistent feed distribution. Early automated systems were based on timer mechanisms that released feed at predefined intervals. While these systems improved consistency compared to manual feeding, they lacked adaptability to changing conditions such as fish size, appetite, and environmental factors.

Recent advancements have introduced sensor-based and weight-based feeding systems that improve accuracy and efficiency. Load cell-based systems, for example, measure the exact amount of feed dispensed, ensuring precise control over feeding quantities. Some systems also incorporate environmental data, such as temperature and dissolved oxygen levels, to adjust feeding schedules dynamically.

Although these developments represent a significant improvement over traditional methods, most automated feeding systems operate independently and are not integrated with other farm management components. This limits their ability to contribute to a comprehensive decision-making process. There remains a need for feeding systems that can operate as part of a larger integrated platform, where feeding decisions are influenced by real-time environmental conditions and overall farm performance.

2.4 Fish Mortality Detection Using Computer Vision

Fish mortality detection is a crucial aspect of aquaculture management, as delayed identification of dead or unhealthy fish can lead to the spread of disease and significant economic losses. Traditional methods rely on visual inspection by farmers, which is time-consuming and often ineffective in detecting early signs of mortality, especially in large or densely populated ponds.

Computer vision and machine learning techniques have emerged as promising solutions for automating fish health monitoring and mortality detection. These systems use cameras to capture images or video streams of the aquatic environment and apply image processing algorithms to analyze fish behavior and appearance. Machine learning models, particularly deep learning algorithms such as convolutional neural networks, have been used to classify fish as healthy, unhealthy, or dead based on visual features.

In addition to static image classification, behaviour-based detection methods have been explored to identify abnormal movement patterns. For example, fish that exhibit unusual floating behaviour or reduced movement may indicate health issues or impending mortality. Optical flow techniques have been used to analyze motion patterns and detect such anomalies in real time.

While these approaches demonstrate high potential, many existing systems face limitations related to environmental conditions such as water turbidity, lighting variations, and camera placement. Moreover, most solutions are developed as independent modules without integration with other farm management systems. This limits their effectiveness in providing comprehensive decision support, as mortality events are not directly linked to environmental conditions or operational factors.

2.5 Water Quality Monitoring and Intelligent Maintenance

Water quality is one of the most important factors affecting fish health and survival. Parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and carbon dioxide must be maintained within specific ranges to ensure a stable aquatic environment. Even slight deviations can cause stress, disease, or mortality in fish populations.

Traditional water quality monitoring methods involve periodic testing and manual observation, which are insufficient for detecting rapid changes. IoT-based monitoring systems have improved this process by enabling continuous measurement and real-time data transmission. Sensors are used to collect environmental data, which is then analyzed to identify abnormal conditions.

Most existing systems rely on threshold-based alerts, where notifications are generated when parameter values exceed predefined limits. While this approach is useful for detecting immediate issues, it does not provide deeper insights into long-term trends or underlying causes. As a result, farmers may struggle to determine appropriate actions based solely on raw data.

To overcome these limitations, advanced systems have introduced the concept of Condition-Based Maintenance. This approach uses real-time data and trend analysis to determine when maintenance activities should be performed. Instead of following fixed schedules, maintenance tasks are generated based on the actual condition of the system. For example, sustained high turbidity levels may indicate the need for filter cleaning, while declining dissolved oxygen levels may suggest aeration issues.

Despite its advantages, the application of condition-based maintenance in aquaculture remains limited. Most systems do not integrate monitoring, analysis, and maintenance planning into a single framework. This highlights the need for a more comprehensive solution that combines real-time monitoring with intelligent decision-making.

2.6 Cost Analytics and Financial Management in Aquaculture

Effective financial management is essential for the sustainability of aquaculture operations. Key cost components include feed, electricity, labor, maintenance, and equipment. In small-scale farms, these costs are typically recorded manually, leading to inconsistencies and limited visibility into overall financial performance.

Existing financial management systems, such as spreadsheets and basic accounting tools, provide only static and retrospective analysis. They do not capture real-time variations in resource usage or link operational activities with financial outcomes. This makes it difficult for farmers to identify inefficiencies, control expenses, and plan budgets effectively.

Recent research has explored the integration of IoT-based data with financial analytics to improve cost management. Energy monitoring systems, for example, use sensors to measure electricity consumption and estimate costs based on usage patterns. Similarly, expense tracking systems can be used to record operational costs in a structured manner.

Predictive forecasting techniques, such as ARIMA and exponential smoothing, have been applied in other domains to estimate future costs and resource requirements. These methods analyze historical data to identify trends and generate predictions. However, their application in aquaculture remains limited, particularly in small-scale environments where data availability is restricted.

The lack of integrated systems that combine real-time monitoring, expense tracking, and predictive analytics represents a significant gap in current research. Addressing this gap can provide farmers with valuable insights into cost structures and support more effective decision-making.

2.7 IoT Communication and Cloud Integration

Efficient communication and data management are critical for the successful operation of IoT-based systems. MQTT has become one of the most widely used communication protocols in IoT applications due to its lightweight design and low bandwidth requirements. It operates on a publish-subscribe model, allowing devices to send data to a central broker and enabling multiple clients to receive updates in real time.

Cloud platforms play an essential role in storing, processing, and analyzing data collected from IoT devices. Services such as real-time databases, authentication systems, and API frameworks enable seamless integration between hardware and software components. These platforms also support remote access, allowing users to monitor and control their systems from anywhere.

While many existing solutions successfully implement communication and cloud integration, they often focus primarily on data visualization rather than advanced analytics or decision support. This limits the overall effectiveness of the system, as users are required to interpret data manually.

2.8 Limitations of Existing Systems

The literature review reveals several key limitations in current aquaculture management systems. Most solutions are designed to address specific aspects such as water quality monitoring or feeding automation, without considering the need for integration across multiple functions. This results in fragmented systems that do not provide a complete view of farm operations.

In addition, many systems rely on simple threshold-based alerts and lack advanced analytical capabilities such as trend analysis, predictive forecasting, and condition-based maintenance. The absence of integration between environmental monitoring, fish health analysis, and financial management further limits the ability of farmers to make informed decisions.

Another major limitation is the lack of affordable solutions tailored for small and medium-scale farms. Many advanced systems are designed for large-scale industrial operations and are not accessible to smaller farmers due to high costs and complexity.

2.9 Summary

This chapter reviewed existing research and technologies related to smart aquaculture systems, including IoT-based monitoring, automated feeding, fish mortality detection, water quality management, and cost analytics. While significant progress has been made in individual areas, current systems remain fragmented and lack integration.

The analysis highlights the need for a unified system that combines automation, monitoring, intelligent analysis, and financial management into a single platform. The proposed IoT-Enabled Smart Aquaculture Management System addresses these gaps by integrating multiple modules and providing a comprehensive solution for improving efficiency, reducing risks, and supporting data-driven decision-making in aquaculture.

3 METHODOLOGY

3.1 Overview

This section describes the methodology used to design and implement the IoT-Enabled Smart Aquaculture Management System. The system integrates four main functional components, namely the Automated Feeding System, Fish Mortality Detection Module, Water Quality Monitoring with Smart Maintenance, and Budget Tracking with Cost Analytics. The objective of this methodology is to combine these modules into a single, unified system that supports real-time monitoring, automation, and intelligent decision-making.

The development approach follows a layered architecture, where data is collected at the device level, transmitted through a communication layer, processed in the cloud, and presented to users through an application interface. The system is designed to operate continuously under real-world conditions, ensuring reliability, scalability, and ease of use for small and medium-scale aquaculture farms. Each module contributes specific functionalities while sharing a common infrastructure for communication and data management.

3.2 System Architecture

The overall system is designed using a multi-layered architecture that ensures efficient integration of hardware and software components. The architecture consists of the device layer, communication layer, cloud processing layer, and application layer. This design enables seamless data flow from sensors and devices to the user interface while maintaining system performance and flexibility.

At the device layer, multiple embedded systems are deployed to handle different functionalities. The automated feeding system uses a microcontroller connected to a load cell and motor mechanism to control feed dispensing. The water quality monitoring module uses sensors to measure parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and carbon dioxide. The fish mortality detection module uses a camera to capture real-time images or video streams for analysis.

The communication layer uses MQTT protocol to enable efficient data transmission between devices and the cloud. Devices publish data to specific topics on the MQTT broker, and the backend system subscribes to these topics to receive and process data in real time. This publish-subscribe model ensures low latency and supports multiple devices simultaneously.

The cloud processing layer handles data storage, analytics, and decision-making. It processes incoming data, applies algorithms for monitoring and prediction, and generates alerts or recommendations. The application layer provides a user-friendly interface through a web-based dashboard, allowing users to monitor system status, view analytics, and interact with different modules.

3.3 Hardware Implementation

The hardware components of the system are designed to ensure accurate data collection and reliable operation in aquaculture environments. The ESP32 microcontroller is used as the primary processing unit due to its built-in Wi-Fi capability, low power consumption, and suitability for IoT applications.

In the automated feeding module, a load cell is used to measure the weight of feed dispensed. The load cell is connected to an amplifier module that converts analog signals into digital data, which is processed by the microcontroller. A motor-driven mechanism controls the release of feed, ensuring precise and consistent feeding.

The water quality monitoring module integrates multiple sensors to measure environmental parameters. These include pH sensors for acidity measurement, dissolved oxygen sensors for oxygen levels, temperature sensors for thermal conditions, turbidity sensors for water clarity, ammonia sensors for toxicity detection, and carbon dioxide sensors for gas concentration analysis. These sensors are connected to the microcontroller through analog and digital interfaces, and proper calibration is performed to ensure accuracy.

The fish mortality detection module uses a camera positioned to capture images or video streams of the pond. The captured data is processed either at the edge or in the cloud to detect abnormal fish behavior or identify dead fish. The placement of the camera is optimized to ensure maximum coverage and minimal interference from environmental factors.

All hardware components are enclosed in protective casings to withstand outdoor conditions such as humidity, temperature variations, and water exposure. Power management is also considered to ensure stable operation.

3.4 Data Acquisition and Processing

The system continuously collects data from sensors and devices at predefined intervals. The microcontroller reads raw data from sensors and performs initial preprocessing steps such as noise filtering and signal smoothing. This helps improve data quality and reduces unnecessary fluctuations.

In the feeding module, the load cell data is used to calculate the exact amount of feed dispensed. In the water quality module, sensor readings are converted into meaningful values representing environmental conditions. In the mortality detection module, image data is captured and prepared for analysis.

The processed data is formatted into structured messages and transmitted to the cloud using MQTT. Each data record includes a timestamp and relevant parameter values, enabling accurate tracking and historical analysis. This approach ensures efficient data handling while minimizing communication overhead.

3.5 Automated Feeding Mechanism

The automated feeding system is designed to provide precise and controlled feed distribution. The system operates based on predefined schedules and weight-based control mechanisms. The load cell continuously measures the amount of feed being dispensed, allowing the system to stop the motor once the required quantity is reached.

This approach ensures consistent feeding and reduces feed wastage. The system can also be extended to adjust feeding schedules based on environmental conditions or fish behavior, providing greater flexibility. The integration of the feeding module with the overall system allows feeding decisions to be influenced by real-time data, improving efficiency and productivity.

3.6 Fish Mortality Detection Method

The fish mortality detection module uses computer vision techniques to analyze images or video streams captured from the pond. The system processes visual data to identify patterns associated with fish health and behavior.

Image processing techniques are used to detect fish presence and movement, while machine learning models classify fish as healthy or dead based on visual features. Behavioral analysis is also applied to detect abnormal patterns such as floating or reduced movement, which may indicate health issues.

The detection results are transmitted to the backend system, where alerts are generated if abnormal conditions are identified. This enables early detection of mortality events and allows farmers to take timely action.

3.7 Water Quality Analysis and Smart Maintenance

The water quality monitoring module analyzes sensor data using multiple techniques to identify abnormal conditions and trends. Instead of relying solely on fixed thresholds, the system incorporates trend analysis and persistence detection to improve accuracy.

Threshold-based analysis checks whether parameter values fall within safe ranges. Trend analysis examines changes over time, while persistence analysis evaluates how long abnormal conditions remain. This combination allows the system to distinguish between temporary fluctuations and serious issues.

The system implements a condition-based maintenance approach, where maintenance tasks are generated based on actual conditions rather than fixed schedules. For example, sustained high turbidity may trigger a cleaning task, while low dissolved oxygen may indicate the need for aeration maintenance. These tasks are displayed on the dashboard and help improve operational efficiency.

3.8 Cost Analytics and Forecasting

The cost analytics module combines energy monitoring and expense tracking to provide a comprehensive view of operational costs. Electrical energy consumption is measured using sensors connected to the power supply, and the data is used to calculate energy costs.

In addition, a structured expense tracking system is used to record costs related to feed, labor, maintenance, and other activities. These data sources are combined to calculate total cost and key performance indicators such as cost per fish and cost per production cycle.

Predictive forecasting techniques, such as ARIMA, are applied to historical data to estimate future costs and resource requirements. This allows farmers to plan budgets and make informed decisions. Forecast accuracy is evaluated using standard error metrics to ensure reliability.

3.9 Communication and Data Flow

The system uses MQTT protocol to enable real-time communication between devices and the backend. The microcontrollers act as publishers, sending data to the MQTT broker, while the backend system subscribes to relevant topics to receive updates.

This publish-subscribe model ensures efficient data transmission and supports scalability. The data flow follows a continuous cycle where data is collected, processed, transmitted, stored, analyzed, and presented to the user. This ensures that all modules operate in a synchronized manner.

3.10 Backend Processing and Storage

The backend system is responsible for managing data storage, processing, and analytics. A cloud-based platform is used to store sensor data, expense records, and analysis results. The backend processes incoming data and applies algorithms to generate insights, alerts, and predictions.

APIs are used to enable communication between the backend and the frontend dashboard. Security mechanisms such as authentication and access control are implemented to protect user data. The system is designed to handle real-time updates and large volumes of data efficiently.

3.11 Alert and Decision Support System

The system includes an alert mechanism that notifies users of abnormal conditions and important events. Alerts are generated when sensor values exceed safe limits, when mortality is detected, or when costs deviate from expected levels.

Notifications are delivered through the dashboard and can be extended to other channels such as SMS or email. The system also includes a cooldown mechanism to prevent repeated alerts for the same condition. This ensures that alerts remain meaningful and actionable.

3.12 System Workflow

The system operates through a continuous workflow that integrates all modules. Sensors and devices collect data, which is processed and transmitted to the cloud. The backend analyzes the data and generates insights, alerts, and maintenance tasks.

Users interact with the system through a dashboard, where they can monitor real-time data, view analytics, and respond to alerts. The integration of all components ensures that the system functions as a unified solution rather than separate modules.

3.13 Evaluation Plan

The system is evaluated based on several performance criteria, including accuracy, reliability, responsiveness, and usability. Sensor readings are compared with reference values to verify accuracy, while system responsiveness is measured based on alert generation time.

The effectiveness of the feeding system, mortality detection, water quality monitoring, and cost analytics modules is evaluated through real-world testing. Continuous operation tests are conducted to ensure system stability. User feedback is also considered to evaluate usability and overall performance.

3.14 Summary

This section presented the methodology used to design and develop the proposed system. The approach integrates multiple technologies, including IoT, computer vision, and data analytics, into a unified platform. The methodology ensures efficient data collection, real-time monitoring, intelligent analysis, and user-friendly interaction, providing a strong foundation for implementation and evaluation.

4 SYSTEM DESIGN AND IMPLEMENTATION

4.1 Overview

This section describes the design and implementation of the IoT-Enabled Smart Aquaculture Management System. The system is developed by integrating hardware components, communication technologies, backend services, and user interfaces into a single, unified platform. The implementation focuses on ensuring reliability, scalability, and real-time performance while maintaining affordability for small and medium-scale aquaculture farms.

The design follows a modular approach where each functional component, including automated feeding, fish mortality detection, water quality monitoring, and cost analytics, is implemented as an independent module. These modules are interconnected through a shared communication and data processing framework, enabling seamless interaction and coordinated operation across the entire system.

4.2 Overall System Architecture

The system architecture is designed using a layered model to ensure efficient data flow and clear separation of responsibilities. The architecture consists of five main layers: sensing and device layer, edge processing layer, communication layer, cloud layer, and application layer.

At the sensing and device layer, various sensors and devices are deployed to collect environmental and operational data. These include water quality sensors, load cells for feeding, cameras for mortality detection, and energy monitoring devices. Each device is connected to a microcontroller that manages data collection and initial processing.

The edge processing layer is responsible for handling local data processing and device control. Microcontrollers such as ESP32 are used to perform tasks such as reading sensor data, controlling motors for feeding, and preparing data for transmission. This layer reduces the amount of raw data sent to the cloud and improves system efficiency.

The communication layer uses MQTT protocol to transmit data between edge devices and the backend system. An MQTT broker manages message distribution, ensuring that data is delivered to the appropriate subscribers in real time.

The cloud layer handles data storage, processing, analytics, and system logic. It processes incoming data, generates alerts and maintenance tasks, and supports predictive analytics. The application layer provides a user interface through a web-based dashboard, allowing users to interact with the system.

4.3 Automated Feeding System Implementation

The automated feeding system is implemented using a combination of hardware and software components designed to ensure precise and consistent feed distribution. The core component of this module is a load cell integrated with a microcontroller to measure the weight of feed being dispensed.

The feeding mechanism uses a motor-driven system controlled by the microcontroller. When a feeding operation is triggered, the motor activates and releases feed into the pond. The load cell continuously measures the weight of the feed, and the system stops the motor once the desired quantity is reached. This ensures accurate feed distribution and minimizes wastage.

The feeding schedule can be configured based on user preferences, allowing farmers to define feeding times and quantities. The system also supports manual control through the dashboard, enabling users to initiate feeding operations remotely. The integration of the feeding system with real-time data allows for future enhancements where feeding decisions can be adjusted based on environmental conditions.

4.4 Fish Mortality Detection Implementation

The fish mortality detection module is implemented using computer vision techniques to analyze visual data captured from the aquaculture environment. A camera is positioned strategically to monitor fish behavior and detect abnormalities in real time.

The captured images or video streams are processed using image analysis algorithms. The system applies preprocessing techniques such as noise reduction and frame extraction to prepare the data for analysis. Machine learning models are used to classify fish as healthy or dead based on visual characteristics.

In addition to classification, the system analyzes movement patterns to detect abnormal behavior. Reduced movement or unusual floating patterns are identified as potential indicators of fish health issues. These observations are processed and transmitted to the backend system, where alerts are generated if abnormal conditions are detected.

The implementation ensures that the module operates efficiently under varying environmental conditions, although factors such as lighting and water clarity are considered during system design.

4.5 Water Quality Monitoring System Implementation

The water quality monitoring module is implemented using a set of sensors connected to a microcontroller. These sensors continuously measure key environmental parameters that affect fish health and survival.

The pH sensor measures the acidity or alkalinity of water, while the dissolved oxygen sensor monitors oxygen levels required for fish respiration. Temperature sensors provide information about thermal conditions, and turbidity sensors measure water clarity. Additional sensors are used to detect ammonia and carbon dioxide levels, providing insights into water toxicity and chemical balance.

The microcontroller reads data from these sensors at regular intervals and performs basic preprocessing to improve accuracy. The processed data is then transmitted to the backend system using MQTT. Calibration procedures are applied to ensure reliable sensor performance.

The system is designed to operate continuously, providing real-time monitoring and enabling early detection of environmental changes. The integration of multiple sensors ensures a comprehensive view of water quality conditions.

4.6 Smart Maintenance Reminder Implementation

The smart maintenance module is implemented as a decision-support system that converts sensor data into actionable maintenance tasks. The system uses predefined rules and analytical techniques to determine when maintenance actions are required.

When sensor data indicates abnormal conditions, such as sustained high turbidity or low dissolved oxygen levels, the system generates maintenance tasks. Each task includes a description, priority level, and recommended action. These tasks are stored in the database and displayed on the user dashboard.

The system allows users to acknowledge and complete tasks, providing a mechanism for tracking maintenance activities. This approach ensures that maintenance is performed based on actual system conditions rather than fixed schedules, improving efficiency and reducing unnecessary effort.

4.7 Cost Analytics and Budget System Implementation

The cost analytics module is implemented to provide insights into operational expenses and support financial decision-making. The system collects data related to energy consumption, feed usage, and other operational costs.

Energy monitoring is implemented using sensors that measure electrical parameters such as voltage and current. The collected data is used to calculate energy consumption and estimate costs. Expense tracking is implemented through a structured system where users can record and categorize expenses.

The system integrates these data sources to calculate total costs and generate financial reports. Predictive analytics techniques are applied to historical data to forecast future expenses. This helps farmers plan budgets and optimize resource usage.

The cost analytics module is integrated with other system components, allowing users to understand the relationship between operational activities and financial performance.

4.8 Communication System Implementation

The communication system is implemented using MQTT protocol, which enables efficient and reliable data transmission between devices and the backend system. An MQTT broker is used to manage message distribution and ensure real-time communication.

Each device publishes data to specific topics, and the backend subscribes to these topics to receive updates. This approach allows multiple devices to communicate simultaneously without interfering with each other. The system is designed to handle network interruptions by implementing reconnection mechanisms and data buffering.

The use of MQTT ensures low latency and efficient use of network resources, making it suitable for IoT-based applications.

4.9 Cloud Deployment and Backend Implementation

The backend system is implemented using a cloud-based platform that provides data storage, processing, and real-time synchronization. The system is deployed on a Virtual Private Server using containerization technologies to ensure scalability and flexibility.

Different services, including the MQTT broker, backend APIs, and database, are hosted in separate containers. This modular approach simplifies system management and allows individual components to be updated or scaled independently.

The backend processes incoming data, applies analytical algorithms, and generates alerts and recommendations. APIs are used to enable communication between the backend and the frontend dashboard.

4.10 Database Design

The database is designed to store sensor data, user information, maintenance tasks, alerts, and financial records. The structure includes multiple tables with defined relationships to ensure data consistency and efficient querying.

Sensor data is stored with timestamps to support real-time monitoring and historical analysis. Maintenance tasks and alerts are stored with details such as status and priority. Financial data is organized to enable cost tracking and reporting.

The database design supports real-time updates and ensures that data can be accessed efficiently by the application layer.

4.11 Frontend Dashboard Implementation

The frontend dashboard is implemented as a web-based application that provides users with access to system data and functionalities. The dashboard displays real-time sensor readings, feeding status, alerts, maintenance tasks, and financial analytics.

Data visualization is implemented using charts and graphs to help users understand trends and patterns. The interface allows users to interact with the system, such as triggering feeding operations, acknowledging alerts, and managing tasks.

The design focuses on simplicity and usability, ensuring that users can easily navigate the system and interpret the information provided.

4.12 System Integration

The system integrates all modules into a single platform, ensuring seamless communication and coordinated operation. Data collected from sensors and devices is processed and shared across modules, enabling comprehensive analysis and decision-making.

The integration allows different modules to influence each other. For example, water quality conditions can impact feeding decisions, and operational data can be linked to cost analytics. This interconnected approach enhances the overall effectiveness of the system.

4.13 Summary

This section presented the design and implementation of the IoT-Enabled Smart Aquaculture Management System. The system integrates multiple components, including hardware devices, communication protocols, cloud services, and user interfaces, into a unified platform. The implementation demonstrates how different technologies can be combined to create an efficient and scalable solution for aquaculture management.

5 TESTING AND EVALUATION

5.1 Overview

This section presents the testing procedures and evaluation methods used to validate the performance of the IoT-Enabled Smart Aquaculture Management System. The objective of testing is to ensure that all integrated modules operate accurately, reliably, and efficiently under real-world conditions. Since the system combines multiple functionalities such as automated feeding, fish mortality detection, water quality monitoring, and cost analytics, the evaluation focuses on both individual module performance and overall system integration.

The testing process is designed to verify system functionality, data accuracy, communication reliability, responsiveness, and usability. A combination of controlled testing and real-world simulation is used to evaluate how well the system meets its objectives.

5.2 Testing Strategy

The testing strategy is structured into three main stages, namely unit testing, integration testing, and system-level testing. Each stage plays a specific role in ensuring the overall quality of the system.

Unit testing focuses on verifying the functionality of individual components. Each module, including sensors, microcontroller operations, communication processes, and backend functions, is tested independently to ensure correctness. For example, sensor readings are checked for accuracy, and feeding mechanisms are tested for precise control.

Integration testing evaluates how different modules interact with each other. This includes testing communication between the ESP32 devices, MQTT broker, backend system, and frontend dashboard. The objective is to ensure that data flows correctly between components without errors or delays.

System-level testing involves evaluating the complete system as a whole. This stage simulates real-world conditions to verify that all modules work together seamlessly. It ensures that the system can operate continuously and respond effectively to changing environmental conditions.

5.3 Functional Testing

Functional testing is conducted to verify that all system features operate according to the defined requirements. Each module is tested using predefined scenarios to ensure correct behavior.

In the automated feeding module, tests are performed to verify that the system dispenses the correct amount of feed based on the configured settings. The load cell readings are monitored to ensure accurate measurement, and the motor control mechanism is tested to confirm proper operation.

For the fish mortality detection module, test cases include identifying healthy and dead fish using sample images or video streams. The system is evaluated based on its ability to correctly classify fish and detect abnormal behavior patterns.

The water quality monitoring module is tested by providing controlled sensor inputs and verifying that the system displays accurate readings and generates alerts when values exceed safe limits. Similarly, the cost analytics module is tested by entering expense data and verifying that calculations and reports are generated correctly.

The results confirm that all functional components operate as expected and meet the system requirements.

5.4 Sensor Accuracy Evaluation

Sensor accuracy is a critical factor in ensuring reliable system performance. Each sensor is tested by comparing its readings with standard reference measurements or calibrated instruments.

The pH sensor is evaluated using buffer solutions with known pH values, while the dissolved oxygen sensor is compared with a commercial measurement device. Temperature readings are verified using a digital thermometer, and turbidity is assessed using standard clarity levels.

The ammonia and carbon dioxide sensors are tested under controlled conditions to observe their responsiveness to changes in gas concentration. The results indicate that most sensors operate within an acceptable error range of approximately five percent, which is sufficient for practical applications.

Regular calibration is recommended to maintain accuracy and ensure consistent performance over time.

5.5 Automated Feeding System Evaluation

The performance of the automated feeding system is evaluated based on accuracy, consistency, and reliability. The system is tested by setting different feeding quantities and measuring the actual feed dispensed using the load cell.

The results show that the system is capable of delivering feed with high precision, maintaining a small margin of error. The motor control mechanism operates smoothly, and the system stops feeding once the required quantity is reached.

The evaluation also confirms that the feeding schedule can be executed consistently without manual intervention. This demonstrates the effectiveness of the automated feeding system in reducing feed wastage and improving operational efficiency.

5.6 Fish Mortality Detection Evaluation

The fish mortality detection module is evaluated based on its ability to accurately identify fish conditions using visual data. Test scenarios include both static image classification and real-time video analysis.

The system successfully detects dead fish and identifies abnormal behavior patterns such as reduced movement or floating. However, certain environmental factors such as lighting conditions and water turbidity can affect detection accuracy.

Despite these challenges, the module provides reliable results in controlled conditions and demonstrates its potential for real-world applications. Further improvements can be made by enhancing model training and optimizing image processing techniques.

5.7 Water Quality Monitoring and Analysis Evaluation

The water quality monitoring module is evaluated based on its ability to collect, process, and analyze environmental data. Sensor readings are monitored over time to assess consistency and responsiveness.

The system effectively detects abnormal conditions using threshold-based alerts and trend analysis. For example, sudden drops in dissolved oxygen levels or increases in turbidity are identified and reported in real time.

The persistence-based analysis further improves accuracy by distinguishing between temporary fluctuations and sustained issues. This ensures that alerts are meaningful and reduces the occurrence of false alarms.

5.8 Smart Maintenance System Evaluation

The smart maintenance module is evaluated based on its ability to generate relevant and timely maintenance tasks using real-time data. The system is tested under different simulated conditions to verify whether it correctly identifies situations that require intervention.

When sustained high turbidity levels are introduced, the system generates cleaning recommendations, while prolonged low dissolved oxygen levels trigger aeration-related maintenance tasks. The system also assigns priority levels based on the severity of the condition, allowing users to focus on critical issues first.

The results show that the module successfully converts raw sensor data into actionable maintenance decisions. This confirms the effectiveness of the condition-based maintenance approach in reducing unnecessary manual effort while ensuring timely intervention.

5.9 Cost Analytics and Forecasting Evaluation

The cost analytics module is evaluated based on its ability to track expenses, calculate costs, and generate meaningful financial insights. The system is tested by inputting various expense records and energy consumption data.

The module accurately calculates total operational costs and provides detailed breakdowns of expenses. The forecasting component is evaluated using historical data, where predictive models generate estimates for future costs. The predictions show reasonable accuracy when compared with expected values.

This evaluation confirms that the system can support financial planning and decision-making by providing clear insights into cost patterns and future trends.

5.10 Communication and Data Transmission Testing

The communication system is tested to evaluate the reliability and efficiency of data transmission between devices and the backend. The MQTT protocol is analyzed under different network conditions to ensure stable performance.

The system successfully transmits sensor data to the backend with minimal delay. Latency is observed to be low, and data is typically available on the dashboard within a few seconds. Network interruptions are also simulated, and the system demonstrates the ability to reconnect and resume communication without significant data loss.

These results confirm that the communication system is suitable for real-time IoT applications and can maintain reliable performance in practical environments.

5.11 System Performance and Reliability

The overall system performance is evaluated through continuous operation testing. The system is run for extended periods to observe stability and reliability under real-world conditions.

The microcontroller operates without interruption, and the backend system processes data consistently. The dashboard updates in real time, providing accurate and up-to-date information to users. The system also demonstrates resilience to temporary failures, ensuring that data is not lost during short-term disruptions.

The results indicate that the system is capable of maintaining stable operation and can be deployed in practical aquaculture environments.

5.12 Usability Evaluation

Usability is evaluated based on user interaction with the system and the ease of understanding its features. The dashboard interface is designed to present information in a clear and organized manner, allowing users to quickly interpret data and respond to alerts.

Users can easily monitor environmental conditions, control feeding operations, and manage maintenance tasks through the interface. The simplicity of the design ensures that even users with limited technical knowledge can operate the system effectively.

The evaluation confirms that the system provides a user-friendly experience and supports efficient interaction.

5.13 Summary of Test Results

The overall testing results demonstrate that the system meets its performance objectives. The sensors provide acceptable accuracy, and the automated feeding system operates with high precision. The fish mortality detection module shows reliable performance under controlled conditions, while the water quality monitoring system effectively identifies environmental changes.

The smart maintenance module successfully generates actionable recommendations, and the cost analytics module provides valuable financial insights. Communication between devices and the backend remains stable, and the system performs reliably under continuous operation.

5.14 Summary

This section presented the testing and evaluation of the IoT-Enabled Smart Aquaculture Management System. The results confirm that the system is accurate, reliable, and effective in improving aquaculture management. The integration of multiple modules into a single platform enhances system functionality and provides a comprehensive solution for monitoring, automation, and decision-making.

6 RESULTS AND DISCUSSION

6.1 Overview

This section presents the results obtained from the implementation of the IoT-Enabled Smart Aquaculture Management System and provides a detailed discussion of system performance. The results are based on observations from all integrated modules, including automated feeding, fish mortality detection, water quality monitoring, and cost analytics. The objective is to evaluate how effectively the system improves aquaculture management and supports decision-making in real-world conditions.

6.2 Real-Time Monitoring Results

The system successfully provides continuous real-time monitoring of environmental and operational parameters. Data collected from sensors, devices, and analytical modules is displayed on the dashboard without noticeable delay. Users are able to observe changes in water quality parameters such as pH, temperature, dissolved oxygen, turbidity, ammonia, and carbon dioxide throughout the day.

The results show that environmental conditions vary significantly depending on feeding activity, weather conditions, and aeration processes. For example, turbidity levels tend to increase immediately after feeding due to suspended particles, while dissolved oxygen levels fluctuate based on temperature and aerator usage. These patterns align with expected real-world behavior, confirming the reliability of the monitoring system.

Compared to traditional manual observation, the system provides a more detailed and continuous understanding of aquaculture conditions, enabling better control and management.

6.3 Automated Feeding System Results

The automated feeding system demonstrates accurate and consistent performance. The load cell-based mechanism ensures that the exact quantity of feed is dispensed during each feeding cycle. Observations show that the system maintains a small margin of error, typically within an acceptable range for practical applications.

The feeding schedule operates reliably without manual intervention, and the system responds correctly to both automated and manual commands. The reduction in feed wastage is one of the key outcomes, as the system avoids overfeeding and ensures optimal feed distribution.

The results indicate that the automated feeding module significantly improves feeding efficiency and contributes to better resource management.

6.4 Fish Mortality Detection Results

The fish mortality detection module is able to identify dead fish and detect abnormal behavior patterns using visual data. The system successfully classifies fish conditions based on image analysis and movement detection.

During testing, the system was able to detect scenarios where fish exhibited reduced movement or abnormal floating patterns. These conditions were correctly identified as potential indicators of health issues. Alerts were generated in real time, allowing for quick response.

However, the results also show that environmental factors such as lighting conditions and water turbidity can influence detection accuracy. Despite these challenges, the module provides valuable insights and demonstrates strong potential for practical use.

6.5 Water Quality Monitoring Results

The water quality monitoring system provides accurate and consistent data across all measured parameters. Sensor readings remain stable under normal conditions and reflect expected variations during environmental changes.

The system effectively detects abnormal conditions, such as low dissolved oxygen levels or high turbidity. These conditions are reported in real time, allowing users to take corrective actions. The inclusion of ammonia and carbon dioxide monitoring enhances the system's ability to detect toxic conditions, which are often overlooked in basic monitoring systems.

The results confirm that the system provides a comprehensive view of water quality and supports effective environmental management.

6.6 Smart Maintenance System Results

The smart maintenance module successfully converts sensor data into actionable maintenance tasks. The system generates maintenance recommendations based on real-time conditions rather than fixed schedules.

For example, sustained high turbidity levels trigger cleaning tasks, while low oxygen levels generate recommendations to check aeration systems. The system assigns priority levels to tasks, helping users focus on critical issues.

This approach improves operational efficiency by ensuring that maintenance activities are performed only when necessary. The results demonstrate that condition-based maintenance is more effective than traditional time-based methods.

6.7 Cost Analytics Results

The cost analytics module provides detailed insights into operational expenses. The system accurately calculates energy consumption and integrates expense data to produce comprehensive cost reports.

The forecasting component generates predictions for future expenses based on historical data. These predictions help users plan budgets and manage resources more effectively. The results show that the module can identify cost patterns and highlight areas where efficiency can be improved.

The integration of cost analytics with operational data allows users to understand the relationship between system performance and financial outcomes.

6.8 System Performance Discussion

The overall system performance is stable and reliable. The integration of multiple modules into a single platform ensures that data flows seamlessly between components. The communication system maintains low latency, and the backend processes data efficiently.

One of the key strengths of the system is its ability to combine real-time monitoring with intelligent analysis. This allows users to move beyond simple observation and make informed decisions based on data. The integration of different modules enhances system functionality and provides a comprehensive solution for aquaculture management.

6.9 Impact of Environmental Conditions

Environmental factors such as temperature changes and rainfall have a noticeable impact on system behavior. The system successfully captures these variations and responds appropriately.

For instance, changes in water parameters following environmental events are detected and reported. This demonstrates the importance of continuous monitoring and highlights the system's ability to handle real-world conditions.

6.10 Limitations and Challenges

Despite its effectiveness, the system has certain limitations. Sensor accuracy can be affected by environmental conditions and requires regular calibration. The fish mortality detection module is sensitive to lighting and water clarity, which can influence detection results.

The system also depends on network connectivity for real-time communication. Although basic mechanisms are in place to handle interruptions, prolonged connectivity issues may affect performance.

These limitations indicate areas for future improvement and further development.

6.11 Summary

This section presented the results and discussion of the IoT-Enabled Smart Aquaculture Management System. The findings demonstrate that the system provides accurate monitoring, efficient automation, and meaningful insights. The integration of multiple modules into a unified platform enhances overall performance and supports effective aquaculture management.

7 CONCLUSION AND FUTURE WORK

7.1 Overview

This section presents the overall conclusion of the IoT-Enabled Smart Aquaculture Management System and summarizes the key outcomes achieved through the research. It evaluates how effectively the proposed system addresses the identified challenges in aquaculture and highlights its contribution to improving efficiency, sustainability, and decision-making. In addition, this section discusses the limitations encountered during development and outlines potential directions for future enhancements.

7.2 Conclusion

This research focused on the design and development of an integrated smart aquaculture system that combines automation, monitoring, analytics, and decision support into a single platform. The proposed system successfully addresses the limitations of traditional aquaculture practices, which rely heavily on manual observation, fixed schedules, and fragmented management approaches.

The implementation demonstrates that automated feeding can be achieved with high accuracy using weight-based control mechanisms, reducing feed wastage and improving consistency. The fish mortality detection module shows the ability to identify abnormal behavior and detect mortality events using computer vision techniques, providing early warnings that help reduce losses.

The water quality monitoring module proves effective in continuously measuring critical environmental parameters such as pH, dissolved oxygen, temperature, turbidity, ammonia, and carbon dioxide. The integration of condition-based maintenance allows the system to generate maintenance tasks based on actual conditions, improving efficiency and preventing environmental deterioration.

The cost analytics module provides valuable insights into operational expenses by integrating energy monitoring and expense tracking with predictive forecasting. This enables users to understand cost patterns, plan budgets, and make informed financial decisions.

The overall system demonstrates reliable performance in terms of data accuracy, communication efficiency, and real-time responsiveness. The integration of all modules into a unified platform allows users to monitor and manage aquaculture operations effectively through a single interface.

7.3 Achievements of the Study

The research achieved several important outcomes. A complete IoT-based aquaculture management system was designed and implemented by integrating hardware devices, communication protocols, cloud services, and user interfaces.

The system successfully established real-time monitoring of environmental and operational parameters, enabling continuous observation of aquaculture conditions. The automated feeding mechanism improved feeding accuracy and reduced resource wastage.

The implementation of fish mortality detection introduced a method for monitoring fish health using visual analysis, enhancing the ability to respond to critical situations. The water quality monitoring module provided comprehensive environmental data and supported intelligent maintenance planning through condition-based strategies.

The cost analytics module introduced financial tracking and forecasting capabilities, allowing users to analyze expenses and plan future operations more effectively. The system also demonstrated stable performance under continuous operation, confirming its suitability for practical use.

7.4 Limitations of the Study

Although the system performs effectively, several limitations were identified during development and testing. Sensor accuracy can be influenced by environmental factors such as temperature, humidity, and water conditions, requiring regular calibration to maintain reliability.

The fish mortality detection module is sensitive to variations in lighting and water clarity, which may affect detection accuracy in certain conditions. Improving robustness in diverse environments remains a challenge.

The system relies on network connectivity for real-time data transmission. While basic mechanisms are implemented to handle temporary disruptions, prolonged connectivity issues may impact system performance.

In addition, the predictive capabilities of the cost analytics module are based on historical data and simple forecasting techniques. More advanced models could improve accuracy and provide deeper insights.

The current implementation is limited to a prototype suitable for small and medium-scale aquaculture farms. Scaling the system for large-scale industrial applications would require further optimization and testing.

7.5 Future Work

Several enhancements can be considered to improve the system in future developments. One important direction is the integration of advanced machine learning models for predictive analysis. This would allow the system to forecast environmental changes, fish behavior, and operational costs more accurately.

The fish mortality detection module can be improved by using more robust computer vision models and expanding the training dataset to include diverse environmental conditions. This would enhance detection accuracy and reliability.

Further development of sensor technologies, particularly for ammonia detection, can improve measurement accuracy and system performance. Incorporating more advanced sensors would provide better insights into water quality.

The system can also be extended to support multiple ponds, allowing centralized monitoring and management of large aquaculture operations. This would increase scalability and applicability in commercial environments.

Developing a dedicated mobile application can enhance accessibility and provide real-time notifications through push alerts. Offline capabilities can also be introduced to handle connectivity issues in remote areas.

Another potential improvement is the integration of automated control mechanisms, such as aerators and water pumps, enabling the system to not only monitor conditions but also take corrective actions automatically.

7.6 Final Remarks

The proposed IoT-Enabled Smart Aquaculture Management System demonstrates the potential of combining modern technologies to improve traditional farming practices. By integrating real-time monitoring, automation, intelligent analysis, and financial management, the system provides a comprehensive solution for aquaculture operations.

The study highlights the importance of data-driven decision-making in improving efficiency, reducing risks, and promoting sustainable practices. The system offers a practical and affordable solution that can be adopted by small and medium-scale farmers, contributing to the advancement of smart aquaculture.

The foundation established by this research opens opportunities for further innovation and development in the field, supporting the transition toward more efficient and technology-driven aquaculture systems.

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9 APPENDICIES